

Mosaicing a Wide Geometric Field of View for Effective Interaction in Augmented Reality

Seokhee Jeon*

Haptics and Virtual Reality Laboratory
Department of Computer Science and Engineering
POSTECH, Republic of Korea

Gerard J. Kim†

Digital Experience Laboratory
Department of Computer Science and Engineering
Korea University, Republic of Korea

Keywords: Mosaicing, Augmented Reality, Spatial Awareness

1 INTRODUCTION

An augmented reality (AR) system combines images from the real environment with properly registered virtual objects. As such, AR systems usually employ see-through head mounted displays (HMD). One popular form is to use the video see-through head mounted display in which real images are obtained through a video camera mounted on the HMD positioned to be near, and aligned with the direction of the user's eyes. This configuration can produce a naturally combined imagery in which the physical view of the user (nearly) matches the view into the actual mixed reality space.

In the case that the camera is attached to the user's head, with the ordinary USB cameras, the user is restricted to see only a limited field of view, i.e. only a small part of the interaction space at a time. The user may have to move one's head very often to interact in the space which is often larger than can be covered by the camera's field of view. And virtual objects may disappear from sight if they are not fully contained within the camera's field of view (or if the head is not directed to the corresponding physical object). This problem is particularly serious in desktop AR where the user does not have much freedom to turn one's head.

Such a problem of DTAR can be solved partly by placing a "fixed" camera at a relatively far location or using multiple cameras and increasing the working field of view. However, as the distance between the interaction space and the camera increases, the tracking and tangible marker recognition reliability decreases, and employing multiple cameras introduces significant additional set up, cost, and computational load. This is a trade-off between convenience/usability and cost. In this paper, we investigate the problem of providing a sufficient software field of view. The ideal case would be when the user has the full view of the interaction space. In particular, we propose to generate and provide wide field of view AR imagery by mosaicing it from the smaller fields of moving views. Figure 1 shows the comparison between a nominal AR view (left) and a mosaiced view. We believe that this scheme will improve task performance by the strengthened spatial awareness and reduced cognitive load.

2 MOSAICING THE INTERACTION SPACE

2.1 Overview

Assuming that the interaction space is much larger than what can be covered by the see-through camera, the nominal configuration only shows a small part of the interaction space at a time. The user is likely to lose spatial awareness, for instance, regarding the object layout, both real and virtual. This can put additional cognitive load in carrying out interactive tasks that encompass the whole

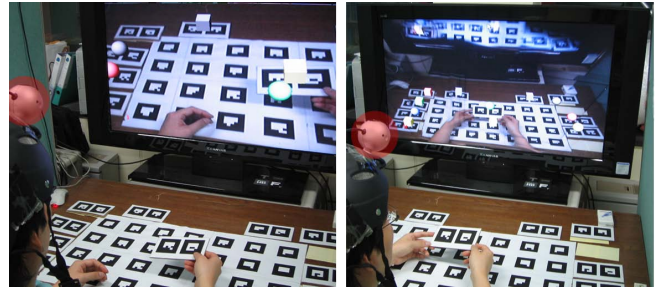


Figure 1: A nominal AR display with a small FOV camera (left) and a mosaiced AR display encompassing most of the interaction space (right). Red circle indicates the camera.

interaction space. The problem can be partial alleviated by providing the display of the full interaction space by stitching imageries from the different parts of the interaction space as the user moves his head/camera during interaction. One of the underlying assumptions is that the views not covered by the current camera direction do not change. This is a reasonable presumption as dynamic interactions normally take place only in a small region where the user is attending at. With a coaxially moving camera on the user's head, user's gaze movement can be estimated using the camera pose tracking data, then the appropriate image can be projected onto the display through homography. However, due to the unstable marker based tracking and image blurring during fast head motions (through which most interaction occurs), additional measures are needed to produce an acceptable image quality in terms of seams among the mosaic image pieces.

2.2 Implementation

The mosaicing algorithm consists of three steps called for each rendering frame. In the first step, the camera's pose with respect to the interaction surface coordination is obtained from the marker-based vision tracker (ARToolkit [1]). In our implementation (for a desktop AR situation), there are 28 markers devoted for this purpose arranged in a grid like fashion on the desk surface at predefined positions (See Figure 1). The ARToolkit tracker finds camera-marker transform matrices of all visible head-tracking markers from an input image frame, and they are merged into one single matrix (M in Figure 2). Inverse of this matrix becomes a camera pose matrix. The Kalman filter is applied to the matrix M to compensate for the jittering in tracking. According to the camera pose matrix, the corresponding image frame is warped using perspective mapping technique as follows. First, four line equations (i.e. lines between the camera and corners of the quadrilateral formed by projecting the rectangular camera image onto the interaction surface) in a camera coordination frame are derived using the intrinsic parameters of the camera. Then, the equations are transformed to the interaction surface coordinate frame by multiply the camera pose matrix. Four intersection points between the interaction surface S and the four lines can be found easily. Using these four point correspondance,

*e-mail: yeager@postech.ac.kr

†e-mail: gjkim@korea.ac.kr, Corresponding Author

the perspective mapping matrix can be calculated.

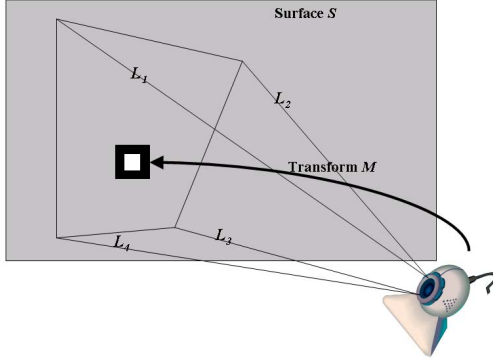


Figure 2: Perspective warping.

The second step is dedicated to reducing stitching errors (and enhancing the mosaic quality) by using a markerless feature tracker (Lukas-Kanade [2]) and an edge smoothing technique. The quality of the mosaiced image largely depends on a quality of tracking data. Even after using the Kalman filter, the instability is still persistent due to e.g. fast movements and other limitations of the AR Toolkit tracker (and a small amount of seams due to this instability is still easily detectable and disturbing to the human eye). To reduce such a problem, tracking errors are traced using markerless feature tracking algorithm as follows. Ten feature points in the current frame are found using the Lucas-Kanade optical flow algorithm. Then, for each feature point, its coordinate on the upfront display (a'_t, b'_t in Figure 3) can be calculated by multiplying its original coordinate (a_t, b_t in Figure 3) by the perspective mapping matrix derived in the first step (M_t in Figure 3). If there is no tracking error, the corresponding two feature points between two consecutive frames ($(a_{t-1}, b_{t-1}), (a_t, b_t)$ in Figure 3) must be the same in the warped coordinate system (i.e. on the upfront display) since these two feature points represent the same feature. In other words, (a'_{t-1}, b'_{t-1}) in Figure 3 must coincide with (a'_t, b'_t) . Extending this idea, we use the sum of Euclidian distance between these two feature pairs as the tracking error for that frame. If the tracking error does not exceed a predefined threshold, this frame is considered a stable frame, and it is used for the mosaicing as a permanent background. Otherwise it is used only at the current frame and discarded. Next procedure for enhancing the mosaic quality is an edge smoothing of the warped image. The edge of the mosaiced image is attenuated by a filter to conceal any remaining seam as much as possible (See Figure 3 bottom).

The virtual objects, registered on the object markers (separate from the head tracking markers), are drawn separately from the background mosaiced image as seen from the virtual camera (instead of the real, using the already known spatial relationship between the real and virtual camera).

In spite of the error correcting procedure, unstable mosaicing (due to tracking error) still may occur when the user moves his/her head (camera) too fast. The tracking error is caused mainly by the motion blur in the captured image (and not being able to recognize any markers). Another drawback of the mosaicing algorithm in its usage is that the assumption of the static background is time to time violated by the moving hand, thus producing images with hands that does not exist at that location anymore. However, overall, the mosaiced image is sufficient to provide sufficiently clear and coherent imagery for interaction and recognizing object layout in the whole interaction space. Our implementation for image mosaicing was done on a 3.2 GHz Pentium 4 PC with a GeForce 7800GTX graphics card.

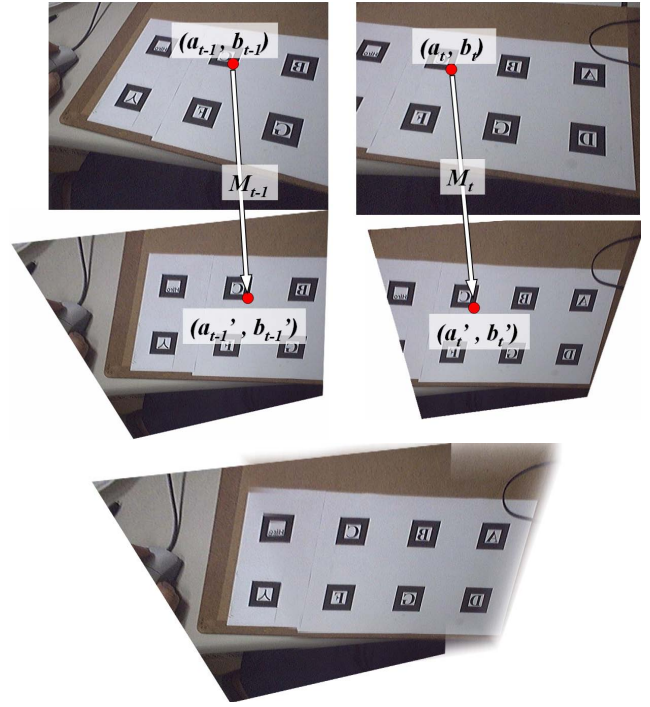


Figure 3: Stitching-error calculation procedure. The left column shows the previous frame and right column, the current frame. The first row shows the original images captured by the camera and the location of feature points in the original image's coordinate system. The second row illustrates the warped images and the warped feature points as well. The bottom picture shows the result of mosaicing two sequential frames after the edge smoothing procedure is carried out.

3 DISCUSSION AND FUTURE WORK

In this paper, we have addressed the usability issue of augmented reality systems. In order to overcome the lack of view into the interaction space in nominal AR systems, we proposed to create and use a mosaiced image as a cost effective and practical solution. One of the obvious problem is that even if the view span is increased by the mosaicing, because of the fixed and limited resolution/size of the display (either HMD or upright display), the content must be viewed in smaller scale. Yang et al. has found out that increasing the geometric field of view can be tolerated to some degree without much degradation in task performance [3]. In our pilot experiment, a comparison of our mosaiced AR display to the nominal set up improved task performance when the task involved manipulation of many tangible props in a relatively wide area. Our future work is to conduct a more formal usability study and address other AR usability issues.

ACKNOWLEDGEMENTS

This research was in part supported by the second BK21 program from the Korean Research Foundation, and the Korea Science and Engineering Foundation (Grant No. R01-2006-000-11142-0).

REFERENCES

- [1] Artoolkit home page. <http://www.hitl.washington.edu/artoolkit/>.
- [2] J. Shi and C. Tomasi. Good features to track. Technical report, Cornell University, 1993.
- [3] U. Yang and G. J. Kim. Increasing the effective egocentric field of view with proprioceptive and tactile feedback. In *IEEE VR*, pages 27–34. IEEE Computer Society, 2004.